

[DC] Beyond Avatars: Designing Bodily Interfaces in Virtual Environments

Xiang Li*

University of Cambridge

ABSTRACT

Our bodies serve as the interface between physical and virtual realities, mediating our perceptions and actions within immersive environments, and enabling the potential for novel experiences or “superpowers.” My PhD research investigates how bodily interfaces can be redesigned to augment user behavior in virtual reality (VR). Specifically, I explore avatar morphologies, transitioning from continuous to discrete forms, such as swarm bodies, to improve the manipulation of multiple objects simultaneously. Additionally, I examine on-body menus, focusing on how the spatial relationship between menu positions and different body landmarks influences user preferences. My ongoing work also addresses microgestures for VR, analyzing their application in text editing and object selection. Moving forward, my research will further investigate control mechanisms for discrete avatars and explore new ways to enhance users’ perception and control of their virtual bodies. Ultimately, my dissertation aims to advance more immersive interactions in virtual environments by leveraging the potential of bodily interfaces.

Index Terms: Bodily interfaces, avatars, spatial interaction, spatial computing, virtual reality, mixed reality.

1 INTRODUCTION

As virtual reality (VR) technologies advance, the boundary between the physical and virtual worlds becomes increasingly blurred. Our bodies, the primary interface for interacting with both realities, now serve as conduits between these dimensions [15, 14]. Mark Hansen, in *Bodies in Code* [6], highlights how our engagement with digital environments is tied to the body’s perception and action, framing the body as a mutable, adaptable medium that shapes human experience. Hansen’s work provides a theoretical foundation for understanding how bodily interfaces in VR not only represent users but also extend their capacities beyond the physical.

The idea of “superpowers” in VR illustrates the transformative potential of bodily interfaces [16]. In immersive environments, users are not limited by their physical bodies but can interact with virtual spaces in ways that amplify their abilities [7]. However, existing VR paradigms—such as controller-based or mid-air interactions—are often too rigid, failing to leverage intrinsic bodily perception for immersion. Prior research shows that user control is essential to create immersive experiences in games and VR, where the congruence between intention and action underpins a sense of agency [4, 3, 1, 2]. When disrupted by rigid control methods, this congruence—and thus immersion—is compromised.

Building on these insights, my research is grounded in the hypothesis that reconfiguring bodily interfaces in VR can enhance engagement and introduce new affordances for interaction, moving beyond simple physical mimicry to offer new control and perception modes. This aligns with the emerging field of human-computer integration, which departs from traditional command models toward shared control between user and machine [5, 13]. In such

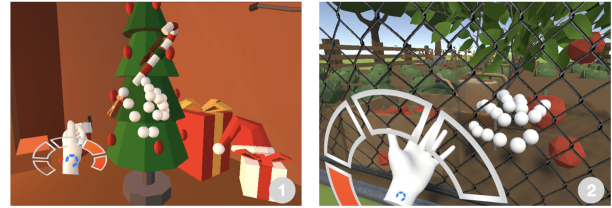


Figure 1: Two open-ended scenarios demonstrating the Swarm Manipulation technique.

systems, autonomy is distributed across the human body and system, fostering shared agency over the interaction [13]. This shared agency allows for a dynamic control style that resonates with users’ cognitive and physical rhythms, moving past the limitations of rigid control [8]. Specifically, I explore how bodily interfaces can leverage the body’s adaptability to extend user capabilities in VR. Key areas include discrete avatar morphologies, such as swarm bodies, which enable simultaneous multi-object manipulation [10, 11], and on-body menus, examining how spatial relationships between menu items and body landmarks affect preference and efficiency [9].

My research also examines microgestures—minimal bodily movements that balance shared agency while addressing VR’s space constraints. Microgestures enable users to control avatars or interact with objects through small, nearly imperceptible movements, reducing the need for large gestures that are often impractical in confined VR setups. This reduction in physical strain enhances VR’s practicality in small spaces and introduces an adaptive, responsive interaction mechanism. For instance, a system might adjust responses based on microgestures influenced by the user’s physiological state or subtle body movements, enhancing immersion by balancing conscious and unconscious control. This adaptive control enables more seamless, intuitive interactions and opens up new possibilities for responsive VR environments.

In conclusion, my research challenges the need for full bodily control in VR by embracing shared agency and exploring alternative control forms—through swarm-based avatars, on-body menus, and microgestures. I aim to contribute to HCI research that reimagines the body’s role in digital environments [12], with new approaches promising richer, more nuanced VR experiences that enhance both control and immersion. Ultimately, my work seeks to deepen our understanding of bodily interfaces to unlock VR’s full potential, fostering dynamic, empowering user experiences.

2 RESEARCH TO DATE

2.1 Swarm Manipulation in VR

As part of my research, I developed a novel swarm manipulation technique in VR (see Figure 1) to address the limitations of traditional interaction methods like virtual hand-based or controller-based approaches [10, 11]. These conventional techniques often struggle with scalability and efficiency when handling multiple objects, as they rely on continuous human-like avatars. To overcome this, we introduced the Swarm Hand, where the user controls a collection of discrete particles, allowing simultaneous manipulation of multiple objects. This swarm-based approach enables users to per-

*e-mail: xl529@cam.ac.uk

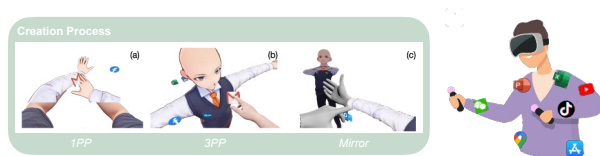


Figure 2: Participants can customize their on-body menu creation process in VR using three perspectives: (a) first-person perspective (1PP), (b) third-person perspective (3PP), and (c) mirror perspective.

form tasks such as selection, rotation, and scaling more fluidly and with greater precision compared to traditional methods. We compared the Swarm Hand to virtual hand and controller-based techniques, and significant improvements were observed in both task efficiency and user satisfaction [10, 11]. Participants completed multi-object manipulation tasks faster and reported a higher sense of control and engagement. The Swarm Hand not only reduced physical effort but also enhanced users' ability to manage complex virtual environments.

2.2 On-Body Menu in VR

Another core part of my research explores the design and evaluation of on-body menus in VR [9]. These menus place icons on various body landmarks, allowing for immersive, direct interactions by utilizing the body as an input platform. Users can access and interact with icons on the arms, torso, and other regions, leveraging proprioception for a more intuitive VR experience.

In our first study, we evaluated three perspective-based creation processes: First-Person (1PP), Third-Person (3PP), and Mirror views. The Mirror perspective resulted in faster menu creation and better icon recall, as users could place and remember icons more effectively with visual confirmation. In contrast, 1PP offered a sense of immersion but was slower and more error-prone due to a limited field of view. Further exploration in our second study examined how users distribute icons across body parts, revealing patterns linked to icon categories. Social Media icons were often placed on the forearms for easy access, while Productivity icons were commonly on the chest or abdomen. These findings suggest that aligning menu layouts with natural body mappings can improve both efficiency and user experience in VR.

3 ONGOING PROJECTS AND FUTURE WORK

Building on prior work with microgestures for VR, two current projects extend the capabilities of precise, subtle input in virtual environments. The first project enhances text editing through small, intuitive gestures, reducing physical strain and improving task accuracy, particularly for text selection and caret navigation, while minimizing dependence on traditional menus. The second project introduces a Bézier curve-based method for object selection, mapped to finger movements, to improve accuracy and disambiguation in dense virtual spaces. This approach integrates on-body interactions to enhance control in situations where traditional ray-casting may be hindered by occlusions.

Future work will broaden the study of bodily interfaces, exploring how different avatar forms, from humanoid to non-humanoid, affect users' sense of ownership and perception in VR. These insights aim to deepen our understanding of VR "superpowers," allowing users to experience interactions that push beyond real-world limitations. Additionally, the mechanism of swarm control in VR offers exciting possibilities. Future studies will examine not only user-swarm interaction but also the self-organizing behavior of swarm particles. Incorporating automation theory could enable autonomous swarm systems that adapt dynamically to user input, ad-

vancing collaborative design, virtual training, and immersive simulations.

ACKNOWLEDGMENTS

Xiang Li is supported by the China Scholarship Council (CSC) International Cambridge Scholarship (No. 202208320092). He sincerely acknowledges his supervisor, Per Ola Kristensson, for his invaluable mentorship and extends his deepest gratitude to all collaborators for their support throughout his PhD journey.

REFERENCES

- [1] J. Bergström, J. Knibbe, H. Pohl, and K. Hornbæk. Sense of Agency and User Experience: Is There a Link? *ACM Transactions on Computer-Human Interaction*, 29(4):28:1–28:22, Mar. 2022. doi: 10.1145/3490493 1
- [2] N. Bianchi-Berthouze, M. Gonzalez-Franco, F. F. Mueller, and M. Sra. Designing computers' control over our bodies (dagstuhl seminar 24232). 2024. 1
- [3] N. Braun, S. Debener, N. Spychala, E. Bongartz, P. Sörös, H. H. Müller, and A. Philipsen. The senses of agency and ownership: a review. *Frontiers in psychology*, 9:535, 2018. 1
- [4] G. Costikyan. Where stories end and games begin. *Game Developer*, 7(9):44–53, 2000. 1
- [5] U. Farooq and J. Grudin. Human-computer integration. *interactions*, 23(6):26–32, 2016. 1
- [6] M. B. Hansen. *Bodies in code: Interfaces with digital media*. Routledge, 2012. 1
- [7] A. Krekhov, S. Cmentowski, K. Emmerich, and J. Krüger. Beyond Human: Animals as an Escape from Stereotype Avatars in Virtual Reality Games. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*, pp. 439–451. ACM, Barcelona Spain, Oct. 2019. doi: 10.1145/3311350.3347172 1
- [8] S.-w. Leigh and P. Maes. Body integrated programmable joints interface. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 6053–6057, 2016. 1
- [9] X. Li, W. He, S. Jin, J. Gugenheimer, P. Hui, H.-N. Liang, and P. O. Kristensson. Investigating creation perspectives and icon placement preferences for on-body menus in virtual reality. *Proc. ACM Hum.-Comput. Interact.*, 8(ISS), Oct. 2024. doi: 10.1145/3698136 1, 2
- [10] X. Li, J.-D. Wang, J. J. Dudley, and P. O. Kristensson. Swarm manipulation in virtual reality. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction, SUI '23*. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3607822.3614519 1, 2
- [11] X. Li, J.-D. Wang, J. J. Dudley, and P. O. Kristensson. Swarm manipulation: An efficient and accurate technique for multi-object manipulation in virtual reality. *Computers & Graphics*, 2024. 1, 2
- [12] F. F. Mueller, R. Byrne, J. Andres, and R. Patibanda. Experiencing the body as play. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2018. 1
- [13] F. F. Mueller, N. Semertzidis, J. Andres, J. Marshall, S. Benford, X. Li, L. Matjeka, and Y. Mehta. Toward understanding the design of intertwined human-computer integrations. *ACM Transactions on Computer-Human Interaction*, 30(5):1–45, 2023. 1
- [14] D. Parisi. Game Interfaces as Bodily Techniques. In R. E. Ferdig, ed., *Handbook of Research on Effective Electronic Gaming in Education*, pp. 111–126. IGI Global, 2009. doi: 10.4018/978-1-59904-808-6.ch007 1
- [15] P. Strohmeier, A. Mottelson, H. Pohl, J. McIntosh, J. Knibbe, J. Bergström, Y. Jansen, and K. Hornbæk. Body-based user interfaces. In *The Routledge Handbook of Bodily Awareness*. Routledge, 2022. Num Pages: 25. 1
- [16] W. Willett, B. A. Aseniero, S. Carpendale, P. Dragicevic, Y. Jansen, L. Oehlberg, and P. Isenberg. Perception! Immersion! Empowerment! Superpowers as Inspiration for Visualization. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):22–32, Jan. 2022. doi: 10.1109/TVCG.2021.3114844 1